TU Clausthal

Exercise – 4

Published on: 01.07.2026

Deadline: 08.07.2026 – 1:59pm

Task(s):

LCA – Part 2 (LCIA + Interpretation)

Task 1

- Attend the LIVE Session after the lecture on 01.06.2026.
- Submit the following to cocalc, following the instructions given during the interactive exercise period:
  - Lifecycle Impact Assessment

    - Determine at least 3 impact categories, e.g. Climate Change, Water Use, Microplastic Pollution for using your washing machine, based on the LCI results from E03.

    - For each Impact Category, define in the following information:

Impact Category	Category Name
LCI results	Emission(s) per Functional Unit
Characterization model	Mathematical model that describes the environmental mechanism linking LCI results to a specific category indicator.
Category indicator	Quantifiable representation of an impact category
Charecterization factor	Factor derived from a characterization model which is applied to convert an assigned LCI result to the common unit of the category indicator
Category indicator result	Total potential impact of a product system for a specific impact category, expressed in the common unit of the selected category indicator.
Category endpoints	Qualitative or quantitative attributes of the environment, human health, or resources that the impact assessment aims to protect.
Environmental relevance	Infrared radiative forcing is a proxy for potential effects on the climate, depending on the integrated atmospheric heat adsorption caused by emissions and the distribution over time of the heat adsorption.

- For each Impact Category, compute

- (1) the environmental impact based on the functional unit (e.g. kg CO2 eq / 10kg of clothes washed), and similarly,

- (2) impact per wash-cycle and,

- (3) impact per year (based on how often you wash clothes)

- (4) the health impact this emission has on humans.

- LCA Interpretation:

  - Identify the main contributors to these environmental impacts

  - Recommend a changes that can be made to reduce this impact.

  - Write a conclusion of what you learned through this LCA (E03 + E04)

Upload your solutions as a PDF file to cocalc.

Additional Hints:

- To compute the amount of microplastics released / functional unit, try to figure out the average percentage of polyester of your clothing.

- Use the category indicator of mg of microplastic fibres. You can derive a characterization model for yourself, based on research papers such as this one: https://www.nature.com/articles/s41598-021-98836-6. This paper found that the release of microplastics is tied to the size of the load. Since we already are assuming a full load, you can assume that ~50 mg/kg of washable load is released (for each wash cycle). Remember to adjust this based on the percentage of polyester. e.g. If on average your clothes are 20% polyester, then you would only use 20% of your max load to compute the microplastic fibre emission.

- You can also use other (academic) resources to find other characterization models for microplastic fibre emissions.

Questions?: etce-etce@tu-clausthal.de